

Infrastructure Description

AWS Services

Udagram uses the following AWS services:

Service	Purpose
Amazon S3	Hosts the compiled Ionic/Angular frontend as a static website. The same deployment also uses S3 object storage for application image files.
AWS Elastic Beanstalk	Hosts the Node.js/Express backend API. The API is deployed from the compiled backend bundle.
Amazon RDS	Hosts the PostgreSQL database used by Sequelize models for users and feed items.
IAM	Provides deployment credentials used by CircleCI to publish builds to S3 and Elastic Beanstalk.

Deployed Resources

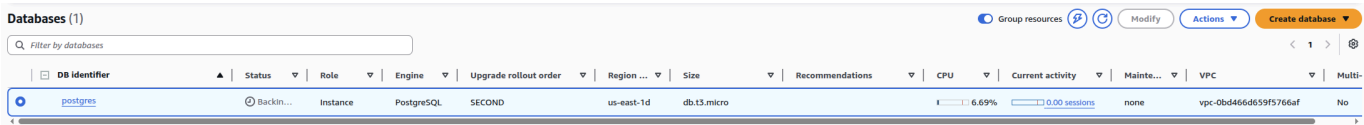
- Frontend S3 bucket: `mj-udacity-udagram-s3`
- Frontend URL: `http://mj-udacity-udagram-s3.s3-website-us-east-1.amazonaws.com`
- API Elastic Beanstalk environment: `Mj-udacity-udagram-api-env`
- API URL: `http://mj-udacity-udagram-api-env.eba-wcfdcn9n.us-east-1.elasticbeanstalk.com/api/v0`
- Database: PostgreSQL on Amazon RDS

Architecture Diagram

See [Architecture.md](#).

Infrastructure Screenshots

RDS



postgres

Summary

DB identifier

postgres

Status

backing-up

CPU

6.50%

Rule

Instance

Current activity

0.00 seconds

Engine

PostgreSQL

Region & AZ

us-east-1d

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Connect using

Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

Endpoints

Use when connecting through any IDE interface.

Internet access gateway

Disabled

Programming language

PostgreSQL

Endpoint type

Instance endpoint

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

1

curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem

2

export RDS_HOSTNAME=postgres.c7zwfj3bcg.us-east-1.rds.amazonaws.com

3

psql "host=\$RDS_HOSTNAME port=5432 dbname=postgres user=postgres sslmode=verify-full sslrootcert=./global-bundle.pem"

Connected compute resources (0)

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

Resource identifier

Resource type

Availability Zone

VPC security group

Compute resource security group

Connected proxy

No connected compute resources

No connected compute resources that were created automatically to display.

Set up EC2 connection

Set up Lambda connection

Security group rules (3)

Filter environments by text, property or value

Security group

Type

Rule

default (sg-04df4f7ed05f9bcdf)

CDR/IP - Inbound

0.0.0.0/0

default (sg-04df4f7ed05f9bcdf)

EC2 Security Group - Inbound

sg-04df4f7ed05f9bcdf

default (sg-04df4f7ed05f9bcdf)

CDR/IP - Outbound

0.0.0.0/0

Elastic Beanstalk

Elastic Beanstalk

Applications

Environments

Change history

Recent environments

Mj-udacity-udagram-api-env

Environment successfully launched.

Environments (1)

Filter environments by text, property or value

Environment name

Health

Application ...

Platform

Domain

Running vers...

Tier name

Date created

Last modified

Mj-udacity-udagram-api-env

OK

mj-udacity-uda...

Node.js 24 run...

Mj-udacity-udagram-api-enve...

-

WebServer

June 13, 2026 1...

June 13, 2026 18:04:46 (UTC+2)

Elastic Beanstalk

Applications

Environments

Change history

Application: mj-udacity-udagram-api

Environment: Mj-udacity-udagram-api-env

Go to environment

Configuration

Events

Deployments

Health & monitoring

Logs

Managed updates

Tags

Recent environments

Mj-udacity-udagram-api-env

Environment successfully launched.

Mj-udacity-udagram-api-env

Environment overview

Health

OK

Domain

Mj-udacity-udagram-api-env-aba-wefdcnrm.us-east-1.elasticbeanstalk.com

Environment ID

e-umpxv3mfe9

Application name

mj-udacity-udagram-api

Platform

Node.js 24 running on 64bit Amazon Linux 2023/6.11.2

Running version

-

Platform state

Supported

Events (19)

Filter events by text, property or value

Time

Type

Details

June 13, 2026 18:04:54 (UTC+2)

INFO

Environment health has transitioned from Pending to Ok. Initialization completed 10 seconds ago and took 3 minutes.

June 13, 2026 18:04:54 (UTC+2)

INFO

Added instance [j-0df9x09fede7e5f0] to your environment.

June 13, 2026 18:04:46 (UTC+2)

INFO

Successfully launched environment: Mj-udacity-udagram-api-env

S3

Amazon S3

Buckets

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1)

Info

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name

AWS Region

Creation date

mj-udacity-udagram-s3

US East (N. Virginia) us-east-1

June 13, 2026, 17:10:55 (UTC+02:00)

mj-udacity-udagram-s3

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)

arn:aws:s3::mj-udacity-udagram-s3

Creation date

June 13, 2026, 17:10:55 (UTC+02:00)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

Bucket ABAC

Edit

Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. [Learn more](#)

ABAC status

Disabled

Default encryption

Edit

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key

When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)

Enabled

Blocked encryption types

Info

Server-Side Encryption with Customer-Provided Keys (SSE-C)

Intelligent-Tiering Archive configurations (0)

View details

Edit

Delete

Create configuration

Enable objects stored in the Intelligent-Tiering storage class to tier-down to the Archive Access tier or the Deep Archive Access tier which are optimized for objects that will be rarely accessed for long periods of time. [Learn more](#)

Find Intelligent-Tiering Archive configurations

Name	Status	Scope	Days until transition to Archive Access t...	Days until transition to Deep Archive A...
No archive configurations No configurations to display.				

Create configuration

Server access logging

Edit

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging

Disabled

AWS CloudTrail data events

You can view and configure CloudTrail data events for Amazon S3 bucket object-level operations in the AWS CloudTrail console.

AWS CloudTrail

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#) ¹
[View analyzer for us-east-1](#)

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) ²

Block all public access
 Off

► Individual Block Public Access settings for this bucket

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) ³

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mj-udacity-udagram-s3/*"
    }
  ]
}
```

Environment Configuration

Environment-specific and secret values are not committed to source code. The API loads configuration from environment variables through its central configuration files under `udagram/udagram-api/src/config`. CircleCI and Elastic Beanstalk provide production values such as:

- `POSTGRES_HOST`
- `POSTGRES_PORT`
- `POSTGRES_DB`
- `POSTGRES_USER`
- `POSTGRES_PASSWORD`
- `AWS_BUCKET`
- `AWS_ACCESS_KEY_ID`
- `AWS_SECRET_ACCESS_KEY`
- `AWS_DEFAULT_REGION`
- `JWT_SECRET`